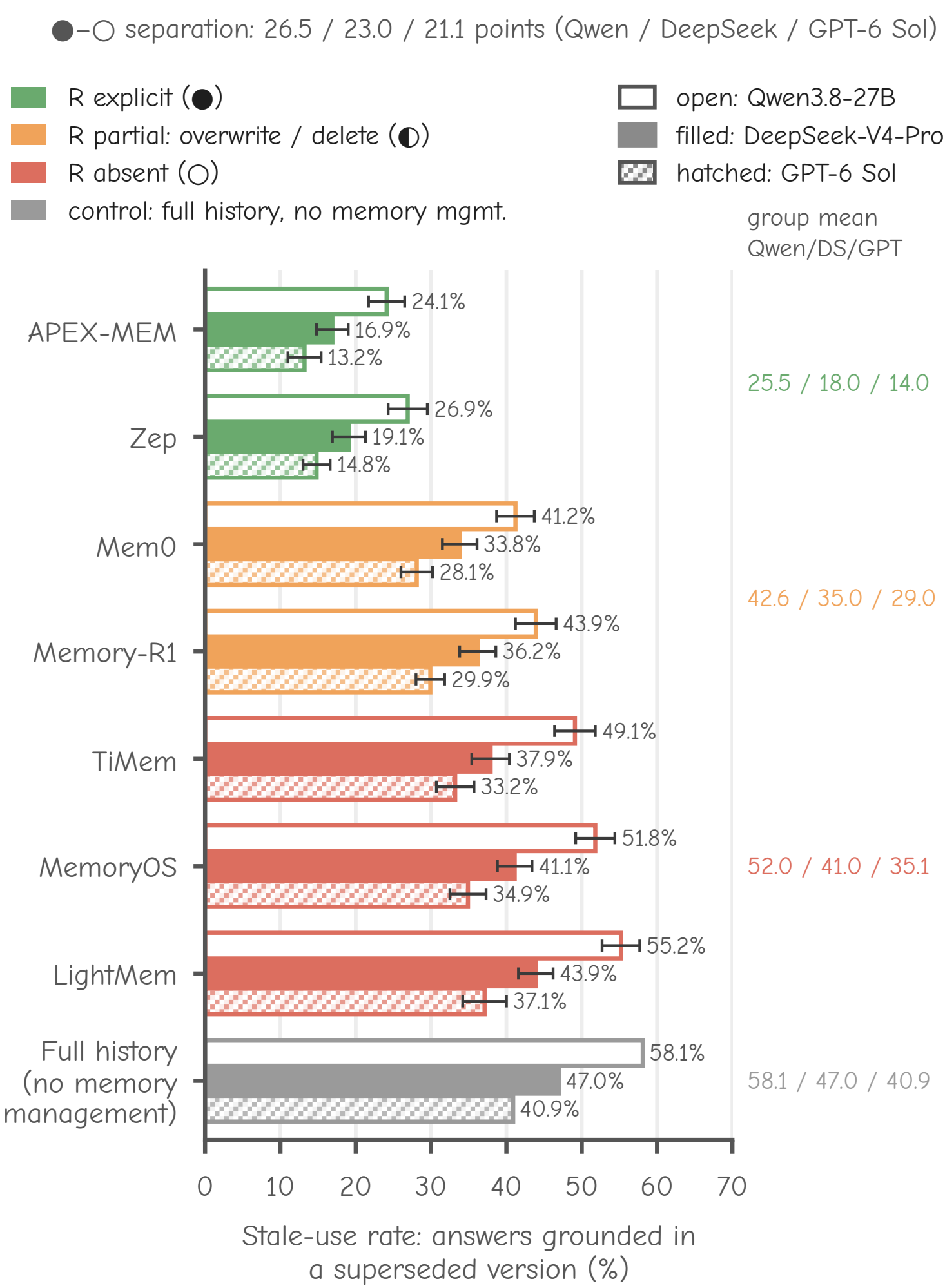
●–○ separation: 26.5 / 23.0 / 21.1 points (Qwen / DeepSeek / GPT-6 Sol)
R explicit (●)
R partial: overwrite / delete (◐)
R absent (○)
control: full history, no memory mgmt.
open: Qwen3.8-27B
filled: DeepSeek-V4-Pro
hatched: GPT-6 Sol
group mean
Qwen/DS/GPT
APEX-MEM
24.1%
16.9%
13.2%
25.5 / 18.0 / 14.0
Zep
26.9%
19.1%
14.8%
Mem0
41.2%
33.8%
28.1%
42.6 / 35.0 / 29.0
Memory-R1
43.9%
36.2%
29.9%
TiMem
49.1%
37.9%
33.2%
MemoryOS
51.8%
41.1%
34.9%
52.0 / 41.0 / 35.1
LightMem
55.2%
43.9%
37.1%
Full history
(no memory
management)
58.1%
47.0%
40.9%
58.1 / 47.0 / 40.9
0
10
20
30
40
50
60
70
Stale-use rate: answers grounded in
a superseded version (%)